# RIYA GUTTIGOLI

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## EDUCATION

**Cornell University, College of Engineering** — **Ithaca, NY**
*M.Eng in Systems Engineering* — *Expected Dec. 2026*
*Bachelor of Science in Mechanical Engineering | Minor: Business* — *May 2026*

## EXPERIENCE

**M.Eng Research Assistant** — **Jan 2026 - Present**
*Cornell Organic Robotics Lab* — *Ithaca, NY*

- Designed and developed a 3-DOF electromechanical test platform to characterize bending behavior of volumetrically manufactured fiber-optic cables, integrating servo-driven motion, custom fixtures, and embedded controls.
- Integrating data acquisition hardware and sensor instrumentation to support experimental characterization and machine-learning based strain prediction.

**Mechanical Engineering Intern - Electromechanical Systems & System Packaging** — **May 2025 - Aug 2025**
*Draper* — *Cambridge, MA*

- Designed adaptive test hardware capable of withstanding vibration environments up to 500 G SRS, applying first-principles analysis, FEA, and GD&T to improve structural performance and manufacturability.
- Developed integrated pneumatic test assembly for high-reliability precision systems, combining mechanical hardware, electrical interfaces, MATLAB data analysis, and LabVIEW instrumentation.
- Reverse-engineered a precision thermal socket component for a rad-hardened IMU and documented it for production.

**Mechanical Engineering Intern - Mechanical & Reliability Systems Team** — **May 2024 - Aug 2024**
*The MITRE Corporation* — *Bedford, MA*

- Modeled and produced 40+ 3D-printed prototypes of electromagnetically-protected enclosures; used SolidWorks, Creo, SpaceClaim, and Icepak for design, defeaturing, and thermal simulation across iterative design cycles.
- Authored two technical white papers on aircraft certification, aerial refueling, and digital-twin integration while collaborating across multidisciplinary engineering teams.

**Mechanical Team Lead** — **Jan 2023 - May 2026**
*Cornell Hyperloop* — *Ithaca, NY*

- Led a multidisciplinary team of 40+ engineers developing a magnetic-levitation transportation system across propulsion, levitation, braking, structures, and electrical integration.
- Led development of scaled MiniPod levitation test platform, including custom electromagnet design, fabrication, characterization, and integration with levitation controls.
- Designed and experimentally validated active electromagnet cooling system using aluminum fins and forced air cooling that reduced operating temperature by 30°C, verified through MATLAB modeling and hardware testing.

## PROJECTS

**ISO New England Electrification Strategy**

- ➢ Modeled long-term electrical demand growth and renewable resource expansion through 2050 using MATLAB to evaluate grid reliability, generation planning, and infrastructure investment strategies.

**Wind Turbine Blade Optimization**

- ➢ Optimized a low-Reynolds-number wind turbine blade using Blade Element Momentum theory and aerodynamic analysis, validating performance through CFD and wind tunnel testing.

**Autonomous Cube Collection Robot**

- ➢ Led mechanical design and subsystem integration for an autonomous competition robot using iterative CAD, rapid prototyping, sensor integration, and experimental testing

**RegenStride**

- ➢ Co-developed a self-healing polymer footwear concept, leading financial modeling, commercialization planning, material cost analysis, and customer discovery.

## TECHNICAL SKILLS

**Design & Systems:** SolidWorks, Creo, Onshape, Fusion360, AutoCAD, SysML
**Simulation:** ANSYS Mechanical, Fluent, Icepak, MATLAB, FEA, CFD
**Programming:** Python, C++, LabVIEW, Arduino
**Manufacturing:** GD&T, Tolerance Analysis, Instrumentation, Data Acquisition, 3D Printing, Machining